On the Non-Uniqueness of Adjacency-Free Colorings of the $n \times n \times n$ Cube

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Abstract

In a companion note we disproved the claim that every six-color, n^2-per-color stickering of an $n \times n \times n$ cube ($n > 1$) must contain two orthogonally adjacent facelets of the same color: a checkerboard construction achieves *zero* monochromatic neighbors. Call such a coloring (six colors, each used exactly n^2 times, no two orthogonally adjacent facelets equal *within* a face) a *proper valid coloring*. Here we settle the natural follow-up: *is such a coloring unique — is there "only one way"?* We prove the answer is **no**, in two independent and strong senses. First, there exist proper valid colorings lying in distinct orbits of the full symmetry group $G = (\text{cube isometries}) \times (\text{color relabelings})$; the *co-occurrence graph* of colors distinguishes them (a perfect matching versus a 6-cycle), so they are not "the same coloring" under any rotation, reflection, or renaming. Second, the raw number $N(n)$ of proper valid colorings grows super-exponentially:

$$N(n) \;\geq\; \left(\frac{(3n^2)!}{(n^2!)^3}\right)^2 \qquad \text{for every } n \geq 2.$$

Hence "exactly one way" fails about as badly as a counting statement can.

1 Setup and the precise question

We keep the model of the companion note. The surface of the cube is the facelet set

$$V = \{(F, i, j) : 1 \leq F \leq 6,\ 1 \leq i, j \leq n\}, \qquad |V| = 6n^2,$$

with faces $F \in \{1, \ldots, 6\}$ and cells (i, j). Two facelets are *neighbors* iff they share a face and are orthogonally adjacent, $|i - i'| + |j - j'| = 1$ (left/right/up/down only; no diagonal, no across-the-edge). The adjacency graph is thus a disjoint union of six $n \times n$ grid graphs. Each face's grid is bipartite with parity classes

$$B = \{(i, j) : i + j \text{ even}\}, \quad W = \{(i, j) : i + j \text{ odd}\}, \quad |B| = \left\lceil \tfrac{n^2}{2} \right\rceil =: a, \ \ |W| = \left\lfloor \tfrac{n^2}{2} \right\rfloor =: b,$$

and $a + b = n^2$ [1].

Definition 1 (Proper valid coloring)**.** A map $c : V \to \{1, \ldots, 6\}$ is a *proper valid coloring* if (i) each color is used exactly n^2 times, and (ii) no two neighbors receive the same color. Let $\mathcal{C}(n)$ denote the set of all such colorings and $N(n) = |\mathcal{C}(n)|$.

The companion note proved $N(n) \geq 1$ (the set is nonempty). The present question is:

(Uniqueness question.) Is there "only one way" — i.e. is $N(n) = 1$, or is the proper valid coloring unique up to the natural symmetries of the cube and of the color set?

Definition 2 (The symmetry group)**.** Let Oct be the group of isometries of the cube (order 48; the 24 rotations and their reflections), acting on V by permuting faces and rigidly moving cells. Let S_6 act by relabeling the six colors. The combined group $G = \mathrm{Oct} \times S_6$ acts on $\mathcal{C}(n)$: $(g, \sigma) \cdot c = \sigma \circ c \circ g^{-1}$. Two colorings are *essentially the same* iff they lie in one G-orbit. "Only one way (up to symmetry)" means $\mathcal{C}(n)$ is a single G-orbit.

We disprove uniqueness in both the literal sense ($N(n) = 1$) and the strong sense (one G-orbit).

2 Two essentially different colorings

We first recall two concrete members of $\mathcal{C}(n)$. Both color every face as a two-color checkerboard, so each is automatically proper: within a face every edge joins B to W, and we always give B and W different colors.

Definition 3 (Construction A: paired faces)**.** Pair the faces as $\{F_1, F_2\}, \{F_3, F_4\}, \{F_5, F_6\}$ and the colors as $\{1, 2\}, \{3, 4\}, \{5, 6\}$. On the pair $\{F_{2k-1}, F_{2k}\}$ with colors $\{2k-1, 2k\}$ set

$$F_{2k-1} : \ B \mapsto 2k-1, \ W \mapsto 2k, \qquad F_{2k} : \ B \mapsto 2k, \ W \mapsto 2k-1.$$

Definition 4 (Construction B: cyclic faces)**.** Arrange the faces in a cycle $F_1, \ldots, F_6$ and set, with indices mod 6,

$$F_i : \quad B \mapsto i, \qquad W \mapsto (i \bmod 6) + 1.$$

Explicitly $F_1 = (1 \mid 2), F_2 = (2 \mid 3), F_3 = (3 \mid 4), F_4 = (4 \mid 5), F_5 = (5 \mid 6), F_6 = (6 \mid 1)$, writing $(x \mid y)$ for "$B \mapsto x, \ W \mapsto y$".

Lemma 1 (Both are proper valid colorings)**.** *For every $n \geq 2$, constructions A and B lie in $\mathcal{C}(n)$.*

Proof. Proper. Each face uses two distinct colors, one on B and one on W; every edge is a B–W edge, so its endpoints differ. *Valid (counts).* In A, color $2k-1$ occurs on $F_{2k-1}.B$ (a cells) and $F_{2k}.W$ (b cells), total $a + b = n^2$; likewise color $2k$. In B, color i occurs on $F_i.B$ (a cells) and on the unique face whose W-image is i, namely F_{i-1} (since $((i-1) \bmod 6) + 1 = i$), contributing b cells; total $a + b = n^2$. Note the count $a + b = n^2$ is independent of the parity of n. □

Definition 5 (Co-occurrence graph)**.** For $c \in \mathcal{C}(n)$ let $H(c)$ be the graph on vertex set $\{1, \ldots, 6\}$ (the colors) with an edge $\{x, y\}$ whenever colors $x \neq y$ both appear on some common face.

Lemma 2 (H is a G-invariant up to isomorphism)**.** *If $c' = (g, \sigma) \cdot c$ then $H(c') \cong H(c)$; in fact σ is a graph isomorphism $H(c) \to H(c')$.*

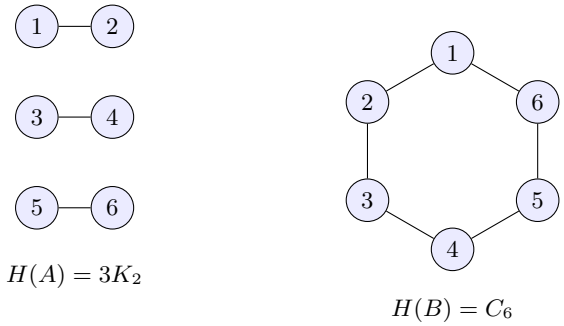
Proof. A cube isometry g permutes the faces but does not change which *pairs of colors* share a face: x, y share face F under c iff they share face $g(F)$ under $c \circ g^{-1}$. Thus the edge set is unchanged by the Oct part. Relabeling colors by σ sends the pair $\{x, y\}$ to $\{\sigma(x), \sigma(y)\}$, so H is carried to its image under σ, i.e. $\sigma : H(c) \to H(c')$ is an isomorphism. Hence the isomorphism type of H is constant on each G-orbit. □

Theorem 1 (Non-uniqueness up to symmetry)**.** *For every $n \geq 2$, constructions A and B are proper valid colorings lying in distinct G-orbits. Consequently $\mathcal{C}(n)$ is* not *a single orbit: there is more than one essentially different adjacency-free coloring.*

Proof. By Lemma 1 both are in $\mathcal{C}(n)$. Their co-occurrence graphs are

$$H(A) = \big\{\{1,2\},\{3,4\},\{5,6\}\big\} = 3K_2 \text{ (a perfect matching)}, \qquad H(B) = \big\{\{i, i{+}1\} : i \in \mathbb{Z}_6\big\} = C_6 \text{ (a 6-cycle)}.$$

In $3K_2$ every vertex has degree 1; in C_6 every vertex has degree 2. The degree sequences differ, so $3K_2 \ncong C_6$. By Lemma 2 the isomorphism type of H is a G-invariant, so A and B cannot lie in the same orbit. Hence $\mathcal{C}(n)$ contains at least two orbits. □



$H(A) = 3K_2$ $\qquad$ $H(B) = C_6$

Remark 1. The argument needs no computer search and no case analysis: a single, cheap invariant (the co-occurrence graph) already separates two explicit colorings. One may push it further — realizing additional isomorphism types of H (e.g. a path P_6, or graphs with triangles via three-colored faces) yields still more orbits — but two already disprove uniqueness.

3 How far from unique: a super-exponential lower bound

Non-uniqueness up to symmetry is a qualitative statement. We now show the raw count explodes.

Theorem 2 (Explosive lower bound)**.** *For every integer $n \geq 2$,*

$$N(n) \;\geq\; \left(\frac{(3n^2)!}{(n^2!)^3}\right)^2.$$

Proof. We exhibit this many distinct proper valid colorings using the *disjoint-palette* principle: if, on each face, the colors allowed on B are disjoint from the colors allowed on W, then every B–W edge automatically has distinct endpoints, so the coloring is proper *no matter how* the cells within B and within W are colored.

Fix the following palette assignment, independent of n:

$$\underbrace{F_1, F_2, F_3}_{\text{“type I”}}: \; B \in \{1,2,3\},\; W \in \{4,5,6\}; \qquad \underbrace{F_4, F_5, F_6}_{\text{“type II”}}: \; B \in \{4,5,6\},\; W \in \{1,2,3\}.$$

On every face the B-palette and W-palette are disjoint, so *every* filling is proper.

Group the cells by which palette colors them. The cells eligible for colors $\{1,2,3\}$ are the B-cells of the three type-I faces and the W-cells of the three type-II faces:

$$\#\{\text{group-}\{1,2,3\}\text{ cells}\} = 3a + 3b = 3(a+b) = 3n^2,$$

and symmetrically the group-$\{4,5,6\}$ cells also number $3b + 3a = 3n^2$. (Both equalities hold for every parity of n, since only $a + b = n^2$ is used.) Every cell lies in exactly one group, and the two groups are colored independently.

Within the group-$\{1,2,3\}$ cells (a set of $3n^2$ *distinguishable* positions) assign colors $1,2,3$ so that each is used exactly n^2 times. The number of such assignments is the multinomial coefficient

$$\binom{3n^2}{n^2,\, n^2,\, n^2} = \frac{(3n^2)!}{(n^2!)^3}.$$

Independently, the group-$\{4,5,6\}$ cells admit $\dfrac{(3n^2)!}{(n^2!)^3}$ assignments with each of $4,5,6$ used n^2 times. Any combination yields a coloring in which all six colors are used exactly n^2 times and (by the disjoint-palette principle) no neighbors match — that is, an element of $\mathcal{C}(n)$. Distinct assignments give distinct colorings because facelets occupy fixed, distinguishable positions. Therefore

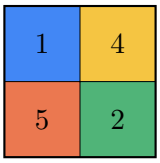
$$N(n) \;\geq\; \left(\frac{(3n^2)!}{(n^2!)^3}\right)^2. \qquad \square$$

Corollary 1. *$N(n) \to \infty$ super-exponentially; already $N(2) \geq \left(\frac{12!}{(4!)^3}\right)^2 = 34650^2 = 1{,}200{,}622{,}500$. In particular $N(n) > 1$ for all $n \geq 2$, so there is certainly not "only one way."*

Proof. By Stirling, $\dfrac{(3m)!}{(m!)^3} \sim \dfrac{3^{3m+1/2}}{2\pi m}$ with $m = n^2$, which grows like 27^{n^2} up to a polynomial factor; squaring keeps it super-exponential in n^2. The value $N(2)$ substitutes $n = 2$, $m = 4$, and $\binom{12}{4,4,4} = 34650$. $\square$

4 Illustration at $n = 2$

For $n = 2$ each face is a 2×2 block (a 4-cycle) with $a = b = 2$. The disjoint-palette scheme of Theorem 2 fills the three type-I faces' B-diagonals and the three type-II faces' W-antidiagonals from $\{1,2,3\}$, and the complementary cells from $\{4,5,6\}$, each color landing exactly $n^2 = 4$ times. One such filling (type-I face F_1 shown using three colors on four cells, demonstrating that faces need not be two-colored):



F_1 (type I): $B \in \{1,2\}$, $W \in \{4,5\}$

Here the two B-cells carry colors $1,2 \in \{1,2,3\}$ and the two W-cells carry $4,5 \in \{4,5,6\}$; the palettes are disjoint, so no edge of the 4-cycle is monochromatic, yet the face shows four colors. Varying which of $1,2,3$ (resp. $4,5,6$) sits in each cell, across all six faces subject to the per-color total of 4, already yields the $34650^2 \approx 1.2 \times 10^9$ colorings counted above — a far cry from a unique solution.

5 Discussion and scope

Verdict. The statement "there is only one way to color the cube so that no two same-colored facelets are adjacent" is **false**, decisively. It fails literally ($N(n) \geq ((3n^2)!/(n^2!)^3)^2 > 1$) and it fails up to the full symmetry group $G = \mathrm{Oct} \times S_6$ (Theorem 1: at least two orbits, witnessed by the non-isomorphic co-occurrence graphs $3K_2$ and C_6).

What drives the multiplicity. Two independent sources. (1) *Combinatorial freedom within a face:* because each face is bipartite, its two parity classes are independent sets, so once a face's B- and W-palettes are kept disjoint, the cells may be filled in any way at all — this is what produces the factorial growth. (2) *Global re-wiring:* even restricting to the simplest two-color-per-face colorings, how the six colors are paired or cycled across faces (matching vs. cycle vs. path) gives genuinely inequivalent structures.

Alternative readings (excluded here). As in the companion note, two stricter models would change the problem. (i) *Counting cross-edge adjacencies* would couple the faces and shrink $\mathcal{C}(n)$, but the disjoint-palette construction can be adapted and multiplicity would persist. (ii) *Restricting to legally reachable scrambles* (fixed centers, permutation/orientation parity on the cube group) is a different and much harder problem; the elementary counting here does not apply to it. Under the "any valid stickering, within-face adjacency" reading requested, Theorems 1 and 2 settle the question completely.

Summary. Not only is the adjacency-free coloring not unique — there are super-exponentially many, falling into several symmetry classes. "Only one way" is as far from the truth as a counting claim can be.

References

[1] R. Diestel, *Graph Theory*, 5th ed., Graduate Texts in Mathematics 173, Springer, 2017. (Bipartite grids; independent sets; proper colorings.)

[2] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001. (Graph isomorphism invariants; degree sequence.)

[3] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, 2nd ed., Cambridge Univ. Press, 2011. (Multinomial coefficients and counting of colorings; group actions on colorings.)